

Electrical Bench Layout + Test Guide

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Purpose: provide a practical, print-friendly game plan for building and validating the electrical module mechanically before final wiring, energization, or permanent enclosure/panel closeout.

Related docs:

- docs/core/ELECTRICAL_48V_ARCHITECTURE.md
- docs/implementation/ELECTRICAL_overview_diagram.md
- docs/implementation/ELECTRICAL_fuse_schedule.md
- docs/plans/PROJECT_build_order_of_operations.md

1) Build intent lock (this guide assumes)

- Architecture remains 48V core + 12V distribution + shore/inverter AC .
- Alternator path baseline is the dedicated 48V secondary alternator path (Mechman + WS500 + APM-48).
- Electrical closet is a two-plane vertical corner build:
 - **Board A:** against bed wall (high-current 48V hardware).
 - **Board B:** against bulkhead (controls, 12V , AC support hardware).
- Closet sits above battery zone.
- Structural mounting intent is through-bolts with washers and lock nuts (no wood screws for permanent component mounting).
- The installed camper shell owns final outer dimensions and service-access geometry; garage/floor layouts are templates only until the two-plane mockup passes in-camper insertion, removal, cable-bend, fuse-access, and panel-access checks.

2) Recommended sequence (ground build -> test fit -> hard mount)

Current owner status (2026-05-11): major electrical components are physically laid out on plywood but not connected, drilled, or energized. The next step is mechanical layout discipline, not live wiring.

Immediate priority:

1. Photograph the current floor layout and label each major component/fuse path.
2. Make 1:1 paper or cardboard templates with terminal sides, cable bend-radius keep-outs, fuse-service access, and wrench access marked.
3. Build/brace the two-plane plywood mockup and move templates onto it.
4. Only after the template layout passes access checks, pilot-drill a small subset of non-energized components.
5. Do not crimp final high-current cables or parallel/energize the 3x 48V bank until the staged test ladder below is ready.

Phase L1: Build the two backer boards on the ground

1. Rip the 1/2" sheet into Board A + Board B to your target envelope.
2. Mark orientation on both boards: TOP , BOTTOM , FRONT , REAR , BED WALL , BULKHEAD .
3. Place boards at 90 degrees on temporary braces/clamps to represent closet geometry.

Exit criteria:

- Two boards stand in correct orientation and are easy to reposition.

Phase L2: Paper layout before drilling

1. Make 1:1 paper templates for each major component.
2. On each template, mark:
 - footprint
 - hole centers
 - cable entry/exit sides
 - keep-out for bend radius + tool access
3. Tape templates to boards and pencil trace candidate positions.
4. Reserve explicit channels:
 - high-current 48V trunk path
 - low-current control/comms path
 - AC route path separated from DC path

Exit criteria:

- No blocked fuse access, no blocked terminals, and no impossible cable bends.

Phase L3: Truck-bed dry fit (before full drilling)

1. Move bare boards (or cardboard stand-ins with same outer dimensions) into truck bed.
2. Validate insertion, rotation, and removal path.
3. Validate service access around door swing/panels.
4. Adjust board size or corner angle now if needed.

Exit criteria:

- Boards can be inserted and removed without forcing or trimming installed components.

Phase L4: Pilot drill + subset mount

1. Pilot drill only (1/8" class) for critical components first.
2. Mount a subset first (example: disconnect, Lynx, shunt, Orion, one AC enclosure).
3. Re-check wrench access, cable swing, and service pull paths.
4. Only after pass, drill final hole sizes and mount remainder.

Exit criteria:

- Critical devices fit and are serviceable with real tools/hands.

Phase L5: Final mounting + board prep

1. Through-bolt permanent component mounts.
2. Add backing washers/plates where loads are concentrated.
3. Seal plywood faces/edges after hole pattern is validated.
4. Label each component position and cable corridor directly on board.

Exit criteria:

- Boards are mechanically complete and labeled for wiring.

3) Suggested zoning by board

Board A (bed-wall plane, high-current focus)

- Place heavier/high-current gear lower when possible for stability and short battery paths.
- Keep these grouped for short 48V runs:
 - main disconnect
 - Lynx
 - shunt
 - major 48V feeds/returns
- Keep alternator branch landing (F-04 path) accessible and clearly labeled.
- Keep Class-T and major fuse service points reachable without removing adjacent equipment.

Board B (bulkhead plane, controls + distribution)

- Place 12V panel, Orion, monitoring/comms, and AC panel hardware here.
- Keep AC hardware grouped and physically separated from DC/control paths.
- Keep control and comms routing high/clean, away from high-current cable bundles.

4) Wiring layout rules (before you crimp anything)

- Keep equal-length parallel battery cables as locked in architecture docs.
 - Keep 48V high-current paths short and direct.
 - Keep AC and DC routing physically separated; cross at 90 degrees only when unavoidable.
 - Use abrasion protection at every edge pass-through.
 - Leave service loops at removable devices (do not strain to exact length).
 - Do not cut final shell-dependent runs yet; only bench-module harnesses at this stage.
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5) Bench test ladder (staged energization)

Use a stop/go ladder; do not skip steps.

T0: Dead checks

- Verify polarity labels and continuity on every cable.
- Verify torque marks applied to all completed terminations.
- Verify fuse IDs/ratings match `docs/implementation/ELECTRICAL_fuse_schedule.md` .
- Before any AC energization, meter the AC system dead: AC-in and AC-out neutrals are isolated, PE/equipment ground is continuous through inlet/enclosure/MultiPlus/receptacle paths, and no fixed downstream neutral-ground bond exists.

Pass when:

- No polarity/continuity mistakes and all terminations documented.

T1: Low-risk controls and comms

- Validate control circuits and monitoring links first (`WS500` control path, comms, panel labeling).
- Confirm `Upfitter #3` control path is treated as low-current signal only.

Pass when:

- Control/monitoring paths behave as expected with no unexplained faults.

T2: 48V core energization (no alternator)

- Bring up house `48V` path first.
- Verify expected voltage at key nodes and no abnormal heat/smell/noise.
- Confirm disconnect behavior and basic inverter/charger DC-side response.

Pass when:

- Stable voltage and normal behavior at all checkpoints.

T2.5: Shore / MultiPlus initial charge validation

- Build the AC-in-only path first: shore source/adapters -> portable EMS -> shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in.
- Keep AC-out branch breakers/loads disconnected for the first charge test.
- Confirm the T0 AC dead-checks are complete before shore energization.
- Connect and charge one `48V` battery at a time through the final-style protected DC path.
- Confirm MultiPlus LiFePO4 charge profile and source current limit before energizing.
- Use the pre-charge procedure before closing the `48V` disconnect into MultiPlus capacitance.
- Leave alternator branch inactive and `F-04` out during shore-charge validation.
- Log starting/resting voltage, charge current, temperature, SOC/monitor readings, abnormal heat/noise/smell, and stop conditions for each battery.

Pass when:

- Each battery charges normally with no unexplained BMS trips, abnormal heat, polarity issues, or AC faults, and the three batteries are close enough to parallel per battery/manual procedure.

T3: 12V distribution validation

- Validate Orion-fed `12V` junction behavior and branch fusing.
- Test representative `12V` loads one at a time, then combined.

Pass when:

- `12V` rail remains stable under expected test loads.

T4: AC path validation

- Validate AC-in and AC-out paths in staged order with small known loads first.
- Reconfirm neutral isolation, PE continuity, no downstream neutral-ground bond, and correct outlet polarity before applying known loads.
- Confirm branch protection behavior and test each GFCI receptacle in its installed branch context.

Pass when:

- AC paths behave correctly in shore and inverter modes, with GFCI behavior verified and no neutral/ground faults found.

T5: Alternator path readiness (pre-engine integration)

- Confirm alternator branch hardware is present/labeled/serviceable.
- Confirm WS500 enable/disable control path labeling and fuse placement (F-15).
- Defer live engine alternator validation until full install context and safety controls are ready.

Pass when:

- Alternator branch is install-ready with shutdown path understood and labeled.

Stop conditions (halt and fix before continuing):

- Any reversed polarity, unexpected voltage, heating at terminals, damaged insulation, nuisance fuse trips, or unclear shutdown behavior.
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6) Garage workday card (print/use)

Use one card per session:

- **Date:**
 - **Board focus:** A / B / Both
 - **Phase:** L1 / L2 / L3 / L4 / L5 / T0-T5
 - **What changed:**
 - **What passed:**
 - **What failed or needs rework:**
 - **Next single action:**
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7) Hold points (do not cross yet)

- Do not finalize shell-dependent cable cuts.
- Do not parallel the three 48V batteries until each has been individually charged/tested and voltage/SOC matching is documented.
- Do not lock roof/solar final terminations.
- Do not permanently close access panels before all staged tests pass.
- Do not use main 48V disconnect as first alternator shutdown method while charging is active; use Upfitter #3 (WS500 disable path) first.